KTB proves neither axiom 4 nor axiom 5

The Open Logic Project contributors

openlogic:normal-modal-logic/axioms-systems:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the load-bearing countermodel figure fig:KTBnot45 (source lines 1578-1599) sits outside the proof environment in the source; it is delivered verbatim as a context span (LANE_BRIEF 15.5/15.7, ren precedent), and the proof cites it via olref which renders as an internal reference to the delivered figure

Editorial note. The label thm:KTBnot45 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Soundness of schemas. If a relation R has one of the listed properties (serial, reflexive, symmetric, transitive, euclidean), every instance of the corresponding axiom schema (D, T, B, 4, 5) is true in every model with that relation (thm:soundschemas, sibling frame-definability unit).. Opens the proof: all instances of T and B are true in every reflexive symmetric model, so by soundness a countermodel need only invalidate one instance of 4 (and of 5).

Source reference. Original theorem thm-soundschemas: sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex, source lines 1566–1566. Named citation invoked at source line(s) 1566: the author's marker \olref[frd][acc]{thm:soundschemas} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. $\mathbf{KTB} \vdash /\mathbf{4}$ *and* $\mathbf{KTB} \vdash /\mathbf{5}$.

Proof. By thm-soundschemas we know that all instances of T and B are true in every reflexive symmetric model (respectively). So by soundness, it suffices to find a reflexive symmetric model containing a world at which some instance

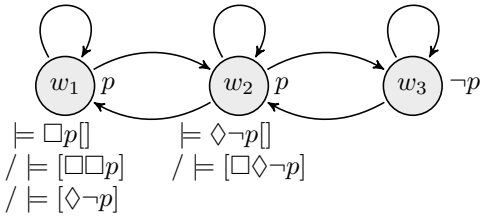

Figure 1: The model for 0.1 .

of 4 fails, and similarly for 5. We use the same model for both claims. Consider the symmetric, reflexive model in 1 . Then $/\models M[\Box p \to \Box\Box p][w_1]$, so 4 fails at w_1. Similarly, $/\models M[\Diamond\neg p \to \Box\Diamond\neg p][w_2]$, so the instance of 5 with $A = \neg p$ fails at w_2. □